Linear Algebra

AICE1004 Mathematics

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Attendance Code from MyEngagement

Figure: Attendance Code

What is Linear Algebra?

Linear Algebra

- It is all about organizing information or data in a **compact and efficient** way such that the follow-up understanding and computation become easy and principled.
- Linear algebra has applications in many fields, including engineering, physics, computer science, economics, and statistics, particularly in areas involving data analysis, **artificial intelligence, machine learning**, optimization, etc.
- What we will cover in the following four weeks?
  - Solving linear systems and its geometrical meaning
  - Vector space, vector, vector operations
  - Matrix and their computation, including addition, multiplication, inverse, determinant, etc
  - Linear independence, matrix diagonalization, eigenvector/eigenvalue
  - Lab: Python programming for matrix computations

Solving linear systems with Gaussian Elimination

Linear systems

Definition A *linear combination* of $x_1, \ldots, x_n$ has the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n$$

where the numbers $a_1, \ldots, a_n \in \mathbb{R}$ are the combination's *coefficients*.

Example This is a linear combination of x, y, and z.

$$(1/4)x + y - z$$

Definition A *linear equation* in the variables $x_1, \ldots, x_n$ has the form $a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = d$ where $d \in \mathbb{R}$ is the *constant*.

An n-tuple $(s_1, s_2, \ldots, s_n) \in \mathbb{R}^n$ is a *solution* of, or *satisfies*, that equation if substituting the numbers $s_1, \ldots, s_n$ for the variables gives a true statement: $a_1s_1 + a_2s_2 + \cdots + a_ns_n = d$. A *system of linear equations*

$$\begin{aligned}
a_{1,1}x_1 + \; a_{1,2}x_2 + \cdots + \; a_{1,n}x_n &= \; d_1 \\
a_{2,1}x_1 + \; a_{2,2}x_2 + \cdots + \; a_{2,n}x_n &= \; d_2 \\
&\vdots \\
a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= \; d_m
\end{aligned}$$

has the solution $(s_1, s_2, \ldots, s_n)$ if that n-tuple is a solution of all of the equations.

Example There are three linear equations in this linear system.

$$\begin{aligned}
(1/4)x + \;\; y - \;\; z &= \;\; 0 \\
x + 4y + 2z &= 12 \\
2x - 3y - \;\; z &= \;\; 3
\end{aligned}$$

Solving a linear system

Example To find the solution of this system

$$\begin{aligned} (1/4)x + y - z &= 0 \\ x + 4y + 2z &= 12 \\ 2x - 3y - z &= 3 \end{aligned}$$

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$$\xrightarrow{4\rho_1} \quad \begin{aligned} x + 4y - 4z &= \ 0 \\ x + 4y + 2z &= 12 \\ 2x - 3y - \ \ z &= \ 3 \end{aligned}$$

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Next use the first row to act on the rows below, eliminating their x terms.

$$\xrightarrow[-2\rho_1+\rho_3]{-\rho_1+\rho_2} \quad \begin{aligned} x + \quad 4y - 4z &= 0 \\ +6z &= 12 \\ -11y + 7z &= 3 \end{aligned}$$

Then swap to bring a y term to the second row.

$$\xrightarrow{\rho_2 \leftrightarrow \rho_3} \quad \begin{aligned} x + \quad 4y - 4z &= 0 \\ -11y + 7z &= 3 \\ 6z &= 12 \end{aligned}$$

Then swap to bring a y term to the second row.

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Now solve the bottom row: $z = 2$.

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Definition In each row of a system, the first variable with a nonzero coefficient is the row's *leading variable*. A system is in *echelon form* if each leading variable is to the right of the leading variable in the row above it, except for the leading variable in the first row, and any rows with all-zero coefficients are at the bottom.

Example

$$
\begin{array}{rcl}
\begin{aligned}
2x - 3y - z + 2w &= -2 \\
x \phantom{{}- 3y - {}} + 3z + 1w &= 6 \\
2x - 3y - z + 3w &= -3 \\
y + z - 2w &= 4
\end{aligned}
&
\xrightarrow[-\rho_1+\rho_3]{(-1/2)\rho_1+\rho_2}
&
\begin{aligned}
2x - 3y - z + 2w &= -2 \\
(3/2)y + (7/2)z \phantom{{}+2w} &= 7 \\
w &= -1 \\
y + z - 2w &= 4
\end{aligned}
\end{array}
$$

Example

$$\begin{array}{rcl} 2x - 3y - z + 2w &=& -2 \\ x \phantom{{}-3y} + 3z + 1w &=& 6 \\ 2x - 3y - z + 3w &=& -3 \\ y + z - 2w &=& 4 \end{array} \xrightarrow[-\rho_1+\rho_3]{(-1/2)\rho_1+\rho_2} \begin{array}{rcl} 2x - 3y - z + 2w &=& -2 \\ (3/2)y + (7/2)z &=& 7 \\ w &=& -1 \\ y + z - 2w &=& 4 \end{array}$$

$$\xrightarrow{(-2/3)\rho_2+\rho_4} \begin{array}{rcl} 2x - 3y - z + 2w &=& -2 \\ (3/2)y + (7/2)z &=& 7 \\ w &=& -1 \\ -(4/3)z - 2w &=& -2/3 \end{array}$$

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$$\xrightarrow{\rho_3 \leftrightarrow \rho_4} \quad \begin{array}{rcl} 2x - 3y - z + 2w &=& -2 \\ (3/2)y + (7/2)z &=& 7 \\ -(4/3)z - 2w &=& -2/3 \\ w &=& -1 \end{array}$$

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2x - 3y - z + 2w &= -2 \\
(3/2)y + (7/2)z &= 7 \\
-(4/3)z - 2w &= -2/3 \\
w &= -1
\end{aligned}$$

The fourth equation says $w = -1$. Substituting back into the third equation gives $z = 2$. Then back substitution into the second and first rows gives $y = 0$ and $x = 1$. The unique solution is $(1, 0, 2, -1)$.

Gauss's Method

?? *Theorem* If a linear system is changed to another by one of these operations

1) an equation is swapped with another
2) an equation has both sides multiplied by a nonzero constant
3) an equation is replaced by the sum of itself and a multiple of another

then the two systems have the same set of solutions.

The proof is in the book.

Definition The three operations from Theorem **??** are the *elementary reduction operations*, or *row operations*, or *Gaussian operations*. They are *swapping*, *multiplying by a scalar* (or *rescaling*), and *row combination*.

Systems without a unique solution

Example This system has no solution.

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + z &= 8 \\ 2x + 3y + 2z &= 13 \end{aligned}$$

On the left the sum of the first two rows equals the third row, while on the right that is not so. So there is no triple of reals that makes all three equations true.

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Gauss' Method makes the inconsistency clear.

$$\xrightarrow[-2\rho_1+\rho_3]{-\rho_1+\rho_2} \begin{array}{rcl} x+y+z &=& 6 \\ y &=& 2 \\ y &=& 1 \end{array} \xrightarrow{-\rho_2+\rho_3} \begin{array}{rcl} x+y+z &=& 6 \\ y &=& 2 \\ 0 &=& -1 \end{array}$$

Example This system has infinitely many solutions.

$$\begin{array}{rcl} -x-y+3z= & 3 \\ x + z= & 3 \\ 3x-y+7z= & 15 \end{array} \xrightarrow[3\rho_1+\rho_3]{\rho_1+\rho_2} \begin{array}{r} -x-y+3z=3 \\ -y+4z=6 \\ -4y+16z=24 \end{array}$$

$$\xrightarrow{-4\rho_2+\rho_3} \begin{array}{r} -x-y+3z=3 \\ -y+4z=6 \\ 0=0 \end{array}$$

Taking $z=0$ gives $(3,-6,0)$ while taking $z=1$ gives $(2,-2,1)$.

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Example It is not the '$0=0$' that counts. This also has infinitely many solutions.

$$\begin{array}{r} x-y+\ z=4 \\ x+y-2z=-1 \end{array} \xrightarrow{-\rho_1+\rho_2} \begin{array}{r} x-\ y+\ z=4 \\ 2y-3z=-5 \end{array}$$

Taking $z=0$ gives the solution $(3/2,-5/2,0)$. Taking $z=-1$ gives $(1,-4,-1)$.

Describing the solution set

Parametrizing

We've seen that this system has infinitely many solutions.

$$\begin{array}{rcl} -x - y + 3z &=& 3 \\ x + z &=& 3 \\ 3x - y + 7z &=& 15 \end{array} \xrightarrow[3\rho_1+\rho_3]{-\rho_1+\rho_2} \xrightarrow{-4\rho_2+\rho_3} \begin{array}{rcl} -x - y + 3z &=& 3 \\ -y + 4z &=& 6 \\ 0 &=& 0 \end{array}$$

We want to describe the solution set.

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We want to describe the solution set.

Use the second row to express y in terms of z as $y=-6+4z$. Now substitute into the first row $-x-(-6+4z)+3z=3$ to express x also in terms of z with $x=3-z$.

This description of the solution set is convenient. Here we pick some z's to show a few of the infinitely many solutions.

z	0	1	2	$-1/2$
solution (x, y, z)	$(3,-6,0)$	$(2,-2,1)$	$(1,2,2)$	$(3.5,-8,-0.5)$

Also, we can tell that $(1,6,3)$ is not a solution without plugging it into the equations because it does not satisfy $x=3-z$.

?? *Definition* In an echelon form linear system the variables that are not leading are *free*. A variable that we use to describe a family of solutions is a *parameter*.

We shall routinely parametrize linear systems with the free variables.

Example This system is already in echelon form.

$$\begin{aligned} 2x + \quad y + z - \quad w &= 5 \\ -y + z + 4w &= 6 \end{aligned}$$

The leading variables are x and y so we will parametrize the solution set with z and w.

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$$\begin{aligned} -2x + y - \;\; z + w &= 3/2 \\ 2z - w &= 1/2 \end{aligned}$$

We parametrize with y and w. The second row gives $z = 1/4 + (1/2)w$. Substituting back into the first row leaves $x = -(7/8) + (1/2)y + (1/4)w$.

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$$\begin{aligned} x &= -(7/8) + (1/2)y + (1/4)w \\ y &= y \\ z &= 1/4 + (1/2)w \\ w &= w \end{aligned}$$

Example

$$
\begin{array}{rcl}
\begin{aligned}
x - y + 2z + 3w &= 14 \\
2x - 2y - z + 2w &= 6 \\
-3z + 2w &= 0
\end{aligned}
&
\xrightarrow{-2\rho_1+\rho_2}
&
\begin{aligned}
x - y + 2z + 3w &= 14 \\
-5z - 4w &= -22 \\
-3z + 2w &= 0
\end{aligned}
\\[1em]
&
\xrightarrow{-(3/5)\rho_2+\rho_3}
&
\begin{aligned}
x - y + 2z + \qquad 3w &= 14 \\
-5z - \qquad 4w &= -22 \\
(22/5)w &= 66/5
\end{aligned}
\end{array}
$$

The leading variables are x, z, and w. We will parametrize with the free variable y.

Example

$$\begin{array}{rcrcrcrcr} x & - & y & + & 2z & + & 3w & = & 14 \\ 2x & - & 2y & - & z & + & 2w & = & 6 \\ & & & - & 3z & + & 2w & = & 0 \end{array} \xrightarrow{-2\rho_1+\rho_2} \begin{array}{rcrcrcrcr} x & - & y & + & 2z & + & 3w & = & 14 \\ & & & - & 5z & - & 4w & = & -22 \\ & & & - & 3z & + & 2w & = & 0 \end{array}$$

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The leading variables are x, z, and w. We will parametrize with the free variable y.

The bottom row gives $w = 3$ and substituting that into the next row up gives $z = 2$. The top equation is $x - y + 2 \cdot 2 + 3 \cdot 3 = 14$ so we have $x = 1 + y$.

$$\begin{aligned} x &= 1 + y \\ y &= y \\ z &= 2 \\ w &= 3 \end{aligned}$$

Matrices and vectors

?? *Definition* An $m \times n$ *matrix* is a rectangular array of numbers with m *rows* and n *columns*. Each number in the matrix is an *entry*.

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$$B = \begin{pmatrix} 1 & -2 & 3 \\ 4 & -5 & 6 \end{pmatrix}$$

because it has 2 rows and 3 columns. The entry in row 2 and column 1 is $b_{2,1} = 4$.

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We denote vectors with an over-arrow or boldface.

Example This column vector has three components.

$$\vec{v} = \begin{pmatrix} -1 \\ -0.5 \\ 0 \end{pmatrix}$$

Example This row vector has three components

$$\vec{w} = \begin{pmatrix} -1 & -0.5 & 0 \end{pmatrix}$$

Example This is the two-component zero vector.

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Vector operations

?? *Definition* The *vector sum* of $\vec{u}$ and $\vec{v}$ is the vector of the sums.

$$\vec{u}+\vec{v}=\begin{pmatrix}u_1\\ \vdots\\ u_n\end{pmatrix}+\begin{pmatrix}v_1\\ \vdots\\ v_n\end{pmatrix}=\begin{pmatrix}u_1+v_1\\ \vdots\\ u_n+v_n\end{pmatrix}$$

?? *Definition* The *scalar multiplication* of the real number r and the vector $\vec{v}$ is the vector of the multiples.

$$r\cdot\vec{v}=r\cdot\begin{pmatrix}v_1\\ \vdots\\ v_n\end{pmatrix}=\begin{pmatrix}rv_1\\ \vdots\\ rv_n\end{pmatrix}$$

Example

$$3\begin{pmatrix}1\\ 2\end{pmatrix}-2\begin{pmatrix}0\\ 1\end{pmatrix}=\begin{pmatrix}3\\ 4\end{pmatrix}$$

Matrix notation for linear systems

Example We can reduce the clerical load in solving this system

$$\begin{aligned} -3x \qquad\quad + 2z &= -1 \\ x - 2y + 2z &= -5/3 \\ -x - 4y + 6z &= -13/3 \end{aligned}$$

by writing it as an *augmented matrix*.

$$\left(\begin{array}{ccc|c} -3 & 0 & 2 & -1 \\ 1 & -2 & 2 & -5/3 \\ -1 & -4 & 6 & -13/3 \end{array}\right) \xrightarrow[-(1/3)\rho_1+\rho_3]{(1/3)\rho_1+\rho_2} \left(\begin{array}{ccc|c} -3 & 0 & 2 & -1 \\ 0 & -2 & 8/3 & -2 \\ 0 & -4 & 16/3 & -4 \end{array}\right)$$

$$\xrightarrow{-2\rho_2+\rho_3} \left(\begin{array}{ccc|c} -3 & 0 & 2 & -1 \\ 0 & -2 & 8/3 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The two nonzero rows give $-3x + 2z = -1$ and $-2y + (8/3)z = -2$.

Parametrizing $-3x + 2z = -1$ and $-2y + (8/3)z = -2$ gives this.

$$\begin{aligned} x &= (1/3) + (2/3)z \\ y &= 1 + (4/3)z \\ z &= z \end{aligned}$$

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We can write the solution set in vector notation.

$$\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2/3 \\ 4/3 \\ 1 \end{pmatrix} z \mid z \in \mathbb{R} \}$$

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This description helps us understand the set of solutions. For instance, each value of z gives a different solution.

$$\begin{array}{r|cccc} z & 0 & 1 & 2 & -1/2 \\ \hline \text{solution } \begin{pmatrix} x \\ y \\ z \end{pmatrix} & \begin{pmatrix} 1/3 \\ 1 \\ 0 \end{pmatrix} & \begin{pmatrix} 1 \\ 7/3 \\ 1 \end{pmatrix} & \begin{pmatrix} 5/3 \\ 11/3 \\ 2 \end{pmatrix} & \begin{pmatrix} 0 \\ 1/3 \\ -1/2 \end{pmatrix} \end{array}$$

Example Reducing this system

$$\begin{aligned} x + 2y - z \phantom{{}+w} &= 2 \\ 2x - y - 2z + w &= 5 \end{aligned}$$

using the augmented matrix notation

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 2 & -1 & -2 & 1 & 5 \end{array}\right) \xrightarrow{-2\rho_1+\rho_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & -5 & 0 & 1 & 1 \end{array}\right)$$

leads to this vector description of the solution set.

$$\{ \begin{pmatrix} 12/5 \\ -1/5 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} z + \begin{pmatrix} -2/5 \\ 1/5 \\ 0 \\ 1 \end{pmatrix} w \mid z, w \in \mathbb{R} \}$$

$$\text{General} = \text{Particular} + \text{Homogeneous}$$

Form of solution sets

Example This system

$$\begin{aligned} x + 2y - z &= 2 \\ 2x - y - 2z + w &= 5 \end{aligned}$$

reduces in this way.

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 2 & -1 & -2 & 1 & 5 \end{array}\right) \xrightarrow{-2\rho_1+\rho_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & -5 & 0 & 1 & 1 \end{array}\right)$$

It has solutions of this form.

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 12/5 \\ -1/5 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} z + \begin{pmatrix} -2/5 \\ 1/5 \\ 0 \\ 1 \end{pmatrix} w \qquad \text{for } z, w \in \mathbb{R}$$

Note that taking $z = w = 0$ shows that the first vector is a particular solution of the system.

Example This system is like the prior one except that is has 0's on the right side. Because of that right side it has a particular solution that is especially simple, namely the zero vector. We say this is the homogeneous system associated with the prior one.

$$\begin{aligned} x + 2y - z \phantom{{}+w} &= 0 \\ 2x - y - 2z + w &= 0 \end{aligned}$$

The same Gauss's Method steps reduce it to echelon form.

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 0 \\ 2 & -1 & -2 & 1 & 0 \end{array}\right) \xrightarrow{-2\rho_1+\rho_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 0 \\ 0 & -5 & 0 & 1 & 0 \end{array}\right)$$

The vector description of the solution set is like the earlier one, but with the zero vector as a particular solution.

$$\{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} z + \begin{pmatrix} -2/5 \\ 1/5 \\ 0 \\ 1 \end{pmatrix} w \mid z, w \in \mathbb{R}\}$$

?? *Definition* A linear equation is *homogeneous* if it has a constant of zero, so that it can be written as $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$.

Homogeneous systems are systems, but . . .

Example This homogeneous system has infinitely many solutions.

$$\begin{aligned} x + y + z &= 0 \\ -2z &= 0 \\ 0 &= 0 \end{aligned}$$

Example This homogeneous system has a unique solution.

$$\begin{aligned} 2x - 3y &= 0 \\ y + z &= 0 \\ 4z &= 0 \end{aligned}$$

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But, a homogeneous system cannot have no solutions. It must have that $x = 0$, . . . is a soution.

?? *Lemma* For any homogeneous linear system there exist vectors $\vec{\beta}_1, \ldots, \vec{\beta}_k$ such that the solution set of the system is

$$\{c_1\vec{\beta}_1 + \cdots + c_k\vec{\beta}_k \mid c_1, \ldots, c_k \in \mathbb{R}\}$$

where k is the number of free variables in an echelon form version of the system.

Example The book has the proof. For the main idea consider this system of homogeneous equations.

$$\begin{aligned} x + y + z + w &= 0 \\ y - z + w &= 0 \end{aligned}$$

Using the bottom equation, express the leading variable y in terms of the free variables, $y = z - w$. Next substitute $x + (z - w) + z + w = 0$ and solve for the leading variable, $x = -2z$.

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$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = z \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \qquad z, w \in \mathbb{R}$$

Recognize the two vectors as the lemma's $\vec{\beta}_1$ and $\vec{\beta}_2$.

?? *Lemma* For a linear system and for any particular solution $\vec{p}$, the solution set equals $\{\vec{p}+\vec{h} \mid \vec{h}$ satisfies the associated homogeneous system$\}$.

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?? *Proof* For set inclusion the first way, that if a vector solves the system then it is in the set described above, assume that $\vec{s}$ solves the system. Then $\vec{s}-\vec{p}$ solves the associated homogeneous system since for each equation index i,

$$\begin{aligned}
&a_{i,1}(s_1-p_1)+\cdots+a_{i,n}(s_n-p_n) \\
&\qquad=(a_{i,1}s_1+\cdots+a_{i,n}s_n)-(a_{i,1}p_1+\cdots+a_{i,n}p_n)=d_i-d_i=0
\end{aligned}$$

where p_j and s_j are the j-th components of $\vec{p}$ and $\vec{s}$. Express $\vec{s}$ in the required $\vec{p}+\vec{h}$ form by writing $\vec{s}-\vec{p}$ as $\vec{h}$.

For set inclusion the other way, take a vector of the form $\vec{p}+\vec{h}$, where $\vec{p}$ solves the system and $\vec{h}$ solves the associated homogeneous system and note that $\vec{p}+\vec{h}$ solves the given system since for any equation index i,

$$\begin{aligned} &a_{i,1}(p_1+h_1)+\cdots+a_{i,n}(p_n+h_n) \\ &\qquad=(a_{i,1}p_1+\cdots+a_{i,n}p_n)+(a_{i,1}h_1+\cdots+a_{i,n}h_n)=d_i+0=d_i \end{aligned}$$

where as earlier p_j and h_j are the j-th components of $\vec{p}$ and $\vec{h}$. QED

?? *Theorem* Any linear system's solution set has the form

$$\{\vec{p} + c_1\vec{\beta}_1 + \cdots + c_k\vec{\beta}_k \mid c_1, \ldots, c_k \in \mathbb{R}\}$$

where $\vec{p}$ is any particular solution and where the number of vectors $\vec{\beta}_1$, ..., $\vec{\beta}_k$ equals the number of free variables that the system has after a Gaussian reduction.

Proof This restates the prior two lemmas. QED

?? *Corollary* Solution sets of linear systems are either empty, have one element, or have infinitely many elements.

Summary: Kinds of Solution Sets

		number of solutions of the homogeneous system	
		one	*infinitely many*
particular solution exists?	*yes*	unique solution	infinitely many solutions
	no	no solutions	no solutions

An important special case is when there are the same number of equations as unknowns.

?? *Definition* A square matrix is *nonsingular* if it is the matrix of coefficients of a homogeneous system with a unique solution. It is *singular* otherwise, that is, if it is the matrix of coefficients of a homogeneous system with infinitely many solutions.

Geometry

We can draw two-unknown equations as lines. Then the three possibilities for solution sets become clear.

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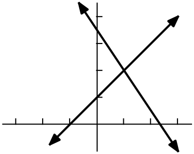
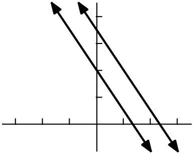
Unique solution



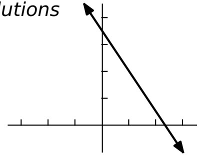
$$\begin{aligned} 3x + 2y &= 7 \\ x - y &= -1 \end{aligned}$$

No solutions



$$\begin{aligned} 3x + 2y &= 7 \\ 3x + 2y &= 4 \end{aligned}$$

Infinitely many solutions



$$\begin{aligned} 3x + 2y &= 7 \\ 6x + 4y &= 14 \end{aligned}$$

This is a nice restatement of the possibilities; the geometry gives us insight into what can happen with linear systems.

Gauss-Jordan reduction

Here is an extension of Gauss's method with some advantages.

Example Start as usual by getting echelon form.

$$\begin{array}{rcl} x + y - z = 2 \\ 2x - y \phantom{{}-z} = -1 \\ x - 2y + 2z = -1 \end{array} \xrightarrow[-1\rho_1+\rho_3]{-2\rho_1+\rho_2} \begin{array}{r} x + y - z = 2 \\ -3y + 2z = -5 \\ -3y + 3z = -3 \end{array}$$

$$\xrightarrow{-1\rho_2+\rho_3} \begin{array}{r} x + y - z = 2 \\ -3y + 2z = -5 \\ z = 2 \end{array}$$

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$$\xrightarrow{-1\rho_2+\rho_3} \quad \begin{array}{rcl} x+\ y-\ z &=& 2 \\ -3y+2z &=& -5 \\ z &=& 2 \end{array}$$

Now, instead of doing back substitution, continue using row operations. First make all the leading entries one.

$$\xrightarrow{(-1/3)\rho_2} \quad \begin{array}{rcl} x+y-\qquad z &=& 2 \\ y-(2/3)z &=& 5/3 \\ z &=& 2 \end{array}$$

Finish by using the leading entries to eliminate upwards, until we can read off the solution.

$$\begin{array}{rcl} x+y-z &=& 2 \\ y-(2/3)z &=& 5/3 \\ z &=& 2 \end{array} \quad \xrightarrow[(2/3)\rho_3+\rho_2]{\rho_3+\rho_1} \quad \begin{array}{rcl} x+y &=& 4 \\ y &=& 3 \\ z &=& 2 \end{array}$$

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$$\xrightarrow{-\rho_2+\rho_1} \quad \begin{array}{rcl} x &=& 1 \\ y &=& 3 \\ z &=& 2 \end{array}$$

Finish by using the leading entries to eliminate upwards, until we can read off the solution.

$$\begin{aligned} x + y - z &= 2 \\ y - (2/3)z &= 5/3 \\ z &= 2 \end{aligned} \quad \xrightarrow[(2/3)\rho_3+\rho_2]{\rho_3+\rho_1} \quad \begin{aligned} x + y &= 4 \\ y &= 3 \\ z &= 2 \end{aligned}$$

$$\xrightarrow{-\rho_2+\rho_1} \quad \begin{aligned} x &= 1 \\ y &= 3 \\ z &= 2 \end{aligned}$$

Using one entry to clear out the rest of a column is *pivoting* on that entry.

Example With this system

$$\begin{aligned} x - y \;\; - 2w &= 2 \\ x + y + 3z + \;\; w &= 1 \\ -y + \;\; z - \;\; w &= 0 \end{aligned}$$

we start by getting echelon form.

$$\xrightarrow{-1\rho_1+\rho_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 2 & 3 & 3 & -1 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right) \xrightarrow{(1/2)\rho_2+\rho_3} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 2 & 3 & 3 & -1 \\ 0 & 0 & 5/2 & 1/2 & -1/2 \end{array}\right)$$

We turn the leading entries to 1's.

$$\xrightarrow[(2/5)\rho_3]{(1/2)\rho_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

Now eliminate upwards.

$$\xrightarrow{-(3/2)\rho_3+\rho_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right) \xrightarrow{\rho_2+\rho_1} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -4/5 & 9/5 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

The final augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -4/5 & 9/5 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

gives the parametrized description of the solution set.

$$\{ \begin{pmatrix} 9/5 \\ -1/5 \\ -1/5 \\ 0 \end{pmatrix} + \begin{pmatrix} 4/5 \\ -6/5 \\ -1/5 \\ 1 \end{pmatrix} w \mid w \in \mathbb{R} \}$$

Gauss-Jordan reduction

This extension of Gauss's Method is the *Gauss-Jordan Method* or *Gauss-Jordan reduction*.

Definition A matrix or linear system is in *reduced echelon form* if, in addition to being in echelon form, each leading entry is a 1 and is the only nonzero entry in its column.

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The cost of using Gauss-Jordan reduction to solve a system is the additional arithmetic. The benefit is that we can just read off the solution set description.

Reduces to is an equivalence

Lemma Elementary row operations are reversible.

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Proof For any matrix A, the effect of swapping rows is reversed by swapping them back, multiplying a row by a nonzero k is undone by multiplying by $1/k$, and adding a multiple of row i to row j (with $i \neq j$) is undone by subtracting the same multiple of row i from row j.

$$A \xrightarrow{\rho_i \leftrightarrow \rho_j} \xrightarrow{\rho_j \leftrightarrow \rho_i} A \qquad A \xrightarrow{k\rho_i} \xrightarrow{(1/k)\rho_i} A \qquad A \xrightarrow{k\rho_i + \rho_j} \xrightarrow{-k\rho_i + \rho_j} A$$

(The third case requires that $i \neq j$.) QED

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(The third case requires that $i \neq j$.) QED

We say that matrices that reduce to each other are equivalent with respect to the relationship of row reducibility. The next result justifies this, using the definition of an equivalence.

Lemma Between matrices, 'reduces to' is an equivalence relation.

The book has the proof. For the intuition, consider this Gauss's method application.

$$\begin{pmatrix}1&2&-1\\2&4&-5\end{pmatrix}\xrightarrow{-2\rho_1+\rho_2}\begin{pmatrix}1&2&-1\\0&0&-3\end{pmatrix}$$

While our experience with Gauss's method leads us to feel that the second matrix in some way "comes after" the first, in fact the two are interreducible. Here are some other 2×3 matrices that are interreducible with those two.

$$\begin{pmatrix}1&2&-1\\0&0&1\end{pmatrix}\quad\begin{pmatrix}2&4&-2\\2&4&-1\end{pmatrix}\quad\begin{pmatrix}1&2&-1\\3&6&-6\end{pmatrix}$$

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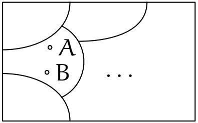
The diagram below shows the collection of all matrices as a box. Inside that box each matrix lies in a class. Matrices are in the same class if and only if they are interreducible. The classes are disjoint — no matrix is in two distinct classes. We have partitioned the collection of matrices into *row equivalence classes*.



Linear Combination Lemma

How Gauss's method acts

Example Consider this reduction.

$$\left(\begin{array}{cc|c}1 & 3 & 5\\ 2 & 4 & 8\end{array}\right) \xrightarrow{-2\rho_1+\rho_2} \left(\begin{array}{cc|c}1 & 3 & 5\\ 0 & -2 & -2\end{array}\right) \xrightarrow{-(1/2)\rho_2} \left(\begin{array}{cc|c}1 & 3 & 5\\ 0 & 1 & 1\end{array}\right) \xrightarrow{-3\rho_2+\rho_1} \left(\begin{array}{cc|c}1 & 0 & 2\\ 0 & 1 & 1\end{array}\right)$$

Denote the matrices as $A \to D \to G \to B$. The steps take us through these row combinations.

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \xrightarrow{-2\rho_1+\rho_2} \begin{pmatrix} \delta_1 = \alpha_1 \\ \delta_2 = -2\alpha_1 + \alpha_2 \end{pmatrix}$$

$$\xrightarrow{-(1/2)\rho_2} \begin{pmatrix} \gamma_1 = \alpha_1 \\ \gamma_2 = \alpha_1 - (1/2)\alpha_2 \end{pmatrix}$$

$$\xrightarrow{-3\rho_2+\rho_1} \begin{pmatrix} \beta_1 = -2\alpha_1 + (3/2)\alpha_2 \\ \beta_2 = \alpha_1 - (1/2)\alpha_2 \end{pmatrix}$$

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So: Gauss's method acts by taking linear combinations of rows.

Linear Combination Lemma

Lemma A linear combination of linear combinations is a linear combination.

For instance, $4x_1 + 5x_2$ and $7x_1 + 8x_2$ are linear combinations, of x_1 and x_2. A linear combination of those

$$3 \cdot (4x_1 + 5x_2) + 6 \cdot (7x_1 + 8x_2)$$

yields another combination, $54x_1 + 63x_2$, of x_1 and x_2.

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Proof Given the set $c_{1,1}x_1 + \cdots + c_{1,n}x_n$ through $c_{m,1}x_1 + \cdots + c_{m,n}x_n$ of linear combinations of the x's, consider a combination of those

$$d_1(c_{1,1}x_1 + \cdots + c_{1,n}x_n) + \cdots + d_m(c_{m,1}x_1 + \cdots + c_{m,n}x_n)$$

where the d's are scalars along with the c's. Distributing those d's and regrouping gives

$$= (d_1c_{1,1} + \cdots + d_mc_{m,1})x_1 + \cdots + (d_1c_{1,n} + \cdots + d_mc_{m,n})x_n$$

which is also a linear combination of the x's. QED

Corollary Where one matrix reduces to another, each row of the second is a linear combination of the rows of the first.

Proof For any two interreducible matrices A and B there is some minimum number of row operations that will take one to the other. We proceed by induction on that number.

In the base step, that we can go from one matrix to another using zero reduction operations, the two are equal. Then each row of B is trivially a combination of A's rows $\vec{\beta}_i = 0 \cdot \vec{\alpha}_1 + \cdots + 1 \cdot \vec{\alpha}_i + \cdots + 0 \cdot \vec{\alpha}_m$.

For the inductive step assume the inductive hypothesis: with $k \geqslant 0$, any matrix that can be derived from A in k or fewer operations has rows that are linear combinations of A's rows. Consider a matrix B such that reducing A to B requires $k+1$ operations. In that reduction there is a next-to-last matrix G, so that $A \longrightarrow \cdots \longrightarrow G \longrightarrow B$. The inductive hypothesis applies to this G because it is only k steps away from A. That is, each row of G is a linear combination of the rows of A.

This states formally what is illustrated in the example showing how Gauss's method acts.

Lemma In an echelon form matrix, no nonzero row is a linear combination of the other nonzero rows.

Proof Let R be an echelon form matrix and consider its non-$\vec{0}$ rows. First observe that if we have a row written as a combination of the others $\vec{\rho}_i = c_1\vec{\rho}_1 + \cdots + c_{i-1}\vec{\rho}_{i-1} + c_{i+1}\vec{\rho}_{i+1} + \cdots + c_m\vec{\rho}_m$ then we can rewrite that equation as

$$\vec{0} = c_1\vec{\rho}_1 + \cdots + c_{i-1}\vec{\rho}_{i-1} + c_i\vec{\rho}_i + c_{i+1}\vec{\rho}_{i+1} + \cdots + c_m\vec{\rho}_m \qquad (*)$$

where not all the coefficients are zero; specifically, $c_i = -1$. The converse holds also: given equation $(*)$ where some $c_i \neq 0$ we could express $\vec{\rho}_i$ as a combination of the other rows by moving $c_i\vec{\rho}_i$ to the left and dividing by $-c_i$. Therefore we will have proved the theorem if we show that in $(*)$ all of the coefficients are 0. For that we use induction on the row number i.

The base case is the first row $i = 1$ (if there is no such nonzero row, so that R is the zero matrix, then the lemma holds vacuously). Let ℓ_i be the column number of the leading entry in row i. Consider the entry of each row that is in column ℓ_1. Equation $(*)$ gives this.

$$0 = c_1 r_{1,\ell_1} + c_2 r_{2,\ell_1} + \cdots + c_m r_{m,\ell_1} \qquad (**)$$

The matrix is in echelon form so every row after the first has a zero entry in that column $r_{2,\ell_1} = \cdots = r_{m,\ell_1} = 0$. Thus equation $(**)$ shows that $c_1 = 0$, because $r_{1,\ell_1} \neq 0$ as it leads the row.

The inductive step is much the same as the base step. Again consider equation $(*)$. We will prove that if the coefficient c_i is 0 for each row index $i \in \{1, \ldots, k\}$ then c_{k+1} is also 0. We focus on the entries from column ℓ_{k+1}.

$$0 = c_1 r_{1,\ell_{k+1}} + \cdots + c_{k+1} r_{k+1,\ell_{k+1}} + \cdots + c_m r_{m,\ell_{k+1}}$$

By the inductive hypothesis $c_1, \ldots c_k$ are all 0 so this reduces to the equation $0 = c_{k+1} r_{k+1,\ell_{k+1}} + \cdots + c_m r_{m,\ell_{k+1}}$. The matrix is in echelon form so the entries $r_{k+2,\ell_{k+1}}, \ldots, r_{m,\ell_{k+1}}$ are all 0. Thus $c_{k+1} = 0$, because $r_{k+1,\ell_{k+1}} \neq 0$ as it is the leading entry.

Example This matrix illustrates.

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

Consider writing the second row as a combination of the others.

$$\begin{pmatrix} 0 & 5 & 6 & 7 \end{pmatrix} = c_1 \cdot (1 \;\; 2 \;\; 3 \;\; 4) + c_3 \cdot (0 \;\; 0 \;\; 0 \;\; 8)$$

We start by looking at the first row. With the matrix in echelon form, the second row's leading entry 5 lies to the right of the first row's 1. The equation of entries in 1's column, the first column, gives that $c_1 = 0$.

$$0 = c_1 \cdot 1 + c_3 \cdot 0$$

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$$0 = c_1 \cdot 1 + c_3 \cdot 0$$

Now for the third row. Again because of echelon from, the third row's leading entry lies to the right of the 5. The equation of entries in the 5's column gives an impossibility.

$$5 = c_3 \cdot 0$$

Summarizing the prior two lemmas: **Gauss's method acts by taking linear combinations of the rows, systematically eliminating any linear relationship among those rows.**

Example In this non-echelon form matrix the third row is the sum of the first and second.

$$\begin{pmatrix} 1 & -1 & 3 \\ 2 & 0 & 4 \\ 3 & -1 & 7 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & -1 & 3 \\ 2 & 0 & 4 \\ 3 & -1 & 7 \end{pmatrix}$$

But after Gauss's method

$$\xrightarrow[-3\rho_1+\rho_3]{-2\rho_1+\rho_3} \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & -2 \\ 0 & 2 & -2 \end{pmatrix} \xrightarrow{-\rho_2+\rho_3} \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

the only linear relationship among the nonzero rows

$$c_1 \cdot \begin{pmatrix} 1 & -1 & 3 \end{pmatrix} = c_2 \cdot \begin{pmatrix} 0 & 2 & -2 \end{pmatrix}$$

is the trivial relationship $c_1 = c_2 = 0$, since the equation of first entries gives that $c_1 = 0$ and then the equation of second entries gives $c_2 = 0$.

Theorem Each matrix is row equivalent to a unique reduced echelon form matrix.

Proof Fix a number of rows m. We will proceed by induction on the number of columns n.

The base case is that the matrix has $n = 1$ column. If this is the zero matrix then its echelon form is the zero matrix. If instead it has any nonzero entries then when the matrix is brought to reduced echelon form it must have at least one nonzero entry, which must be a 1 in the first row. Either way, its reduced echelon form is unique.

For the inductive step we assume that $n > 1$ and that all m row matrices having fewer than n columns have a unique reduced echelon form. Consider an $m \times n$ matrix A and suppose that B and C are two reduced echelon form matrices derived from A. We will show that these two must be equal.

Let $\hat{A}$ be the matrix consisting of the first $n - 1$ columns of A. Observe that any sequence of row operations that bring A to reduced echelon form will also bring $\hat{A}$ to reduced echelon form. By the inductive hypothesis this reduced echelon form of $\hat{A}$ is unique, so if B and C differ then the difference must occur in column n.

Consider a homogeneous system of equations for which A is the matrix of coefficients.

$$\begin{aligned} a_{1,1}x_1 + \ a_{1,2}x_2 + \cdots + \ a_{1,n}x_n &= 0 \\ a_{2,1}x_1 + \ a_{2,2}x_2 + \cdots + \ a_{2,n}x_n &= 0 \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= 0 \end{aligned} \qquad (*)$$

By Theorem One.I.**??** the set of solutions to that system is the same as the set of solutions to B's system

$$\begin{aligned} b_{1,1}x_1 + \ b_{1,2}x_2 + \cdots + \ b_{1,n}x_n &= 0 \\ b_{2,1}x_1 + \ b_{2,2}x_2 + \cdots + \ b_{2,n}x_n &= 0 \\ &\vdots \\ b_{m,1}x_1 + b_{m,2}x_2 + \cdots + b_{m,n}x_n &= 0 \end{aligned} \qquad (**)$$

and to C's.

$$\begin{aligned} c_{1,1}x_1 + \ c_{1,2}x_2 + \cdots + \ c_{1,n}x_n &= 0 \\ c_{2,1}x_1 + \ c_{2,2}x_2 + \cdots + \ c_{2,n}x_n &= 0 \\ &\vdots \\ c_{m,1}x_1 + c_{m,2}x_2 + \cdots + c_{m,n}x_n &= 0 \end{aligned} \qquad (***)$$

With B and C different only in column n, suppose that they differ in row i. Subtract row i of ($***$) from row i of ($**$) to get the equation $(b_{i,n} - c_{i,n}) \cdot x_n = 0$. We've assumed that $b_{i,n} \neq c_{i,n}$ and so we get $x_n = 0$. Thus x_n is not a free variable and so in ($**$) and ($***$) the n-th column contains the leading entry of some row, since in an echelon form matrix any column that does not contain a leading entry is associated with a free variable.

But now, with B and C equal on the first $n-1$ columns, by the definition of reduced echeleon form their leading entries in the n-th column are in the same row. And, both leading entries would have to be 1, and would have to be the only nonzero entries in that column. Therefore $B = C$. QED

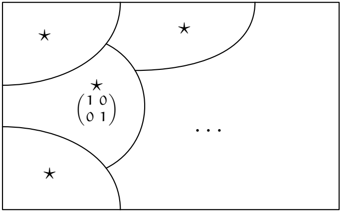
Example This Gauss-Jordan reduction shows that the matrix on the left is row equivalent to the reduced echelon form matrix on the right.

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 3 & 6 & 1 \end{pmatrix} \xrightarrow[-3\rho_1+\rho_3]{-2\rho_1+\rho_2} \xrightarrow{-\rho_2+\rho_3} \xrightarrow{(1/2)\rho_2} \xrightarrow{-\rho_2+\rho_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

By the theorem the matrix on the left is not row equivalent to any of these.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So the reduced echelon form is a canonical form for row equivalence: the reduced echelon form matrices are representatives of the classes.



Example To decide if these two are row equivalent

$$\begin{pmatrix} 3 & 2 & 0 \\ 1 & -1 & 2 \\ 4 & 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 3 & 1 & -2 \\ 6 & 2 & -4 \\ 1 & 0 & 2 \end{pmatrix}$$

use Gauss-Jordan elimination to bring each to reduced echelon form and see if they are equal.

Example To decide if these two are row equivalent

$$\begin{pmatrix}3&2&0\\1&-1&2\\4&1&2\end{pmatrix}\qquad\begin{pmatrix}3&1&-2\\6&2&-4\\1&0&2\end{pmatrix}$$

use Gauss-Jordan elimination to bring each to reduced echelon form and see if they are equal. The results are

$$\xrightarrow[-(4/3)\rho_1+\rho_3]{-(1/3)\rho_1+\rho_2}\ \xrightarrow{-1\rho_2+\rho_3}\ \xrightarrow[-(3/5)\rho_2]{(1/3)\rho_1}\ \xrightarrow{-(2/3)\rho_2+\rho_1}\begin{pmatrix}1&0&4/5\\0&1&-6/5\\0&0&0\end{pmatrix}$$

and

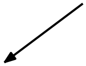
$$\xrightarrow[-(1/3)\rho_1+\rho_3]{-2\rho_1+\rho_2}\ \xrightarrow{\rho_2\leftrightarrow\rho_3}\ \xrightarrow[-3\rho_2]{(1/3)\rho_1}\ \xrightarrow{-(1/3)\rho_2+\rho_1}\begin{pmatrix}1&0&2\\0&1&-8\\0&0&0\end{pmatrix}$$

and therefore the original matrices are not row equivalent.

Vectors in space

Vectors

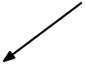
A *vector* is an object consisting of a magnitude and a direction.



A vector can model a displacement.

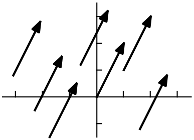
Vectors

A *vector* is an object consisting of a magnitude and a direction.



A vector can model a displacement.

Two vectors with the same magnitude and same direction, such as all of these, are equal.



For instance, each of the above could model a displacement of one over and two up.

Denote the vector that extends from (a_1, a_2) to (b_1, b_2) by

$$\begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \end{pmatrix}$$

so the "one over, two up" vector would be written in this way.

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

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$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

We often picture a vector

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

as starting at the origin. From there $\vec{v}$ extends to (v_1, v_2) and we may refer to it as "the point $\vec{v}$" so that we may call each of these $\mathbb{R}^2$.

$$\{(x_1, x_2) \mid x_1, x_2 \in \mathbb{R}\} \qquad \{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mid x_1, x_2 \in \mathbb{R}\}$$

These definitions extend to higher dimensions. The vector that starts at $(a_1, \ldots, a_n)$ and ends at $(b_1, \ldots, b_n)$ is represented by this column

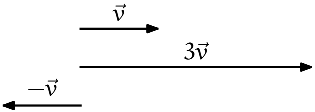
$$\begin{pmatrix} b_1 - a_1 \\ \vdots \\ b_n - a_n \end{pmatrix}$$

and two vectors are equal if they have the same representation. Also, we aren't too careful about distinguishing between a point and the vector which, when it starts at the origin, ends at that point.

$$\mathbb{R}^n = \{ \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \mid v_1, \ldots, v_n \in \mathbb{R} \}$$

Vector operations

Scalar multiplication makes a vector longer or shorter, including possibly flipping it around.



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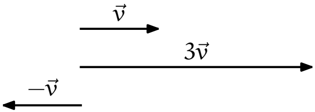
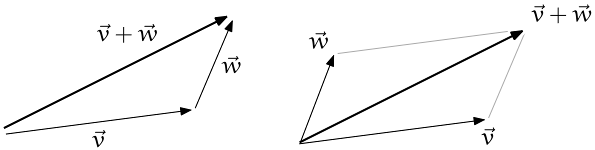


Where $\vec{v}$ and $\vec{w}$ represent displacements, the vector sum $\vec{v}+\vec{w}$ represents those displacements combined.

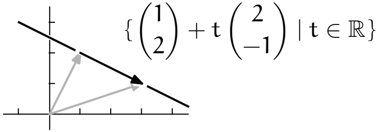


The second drawing shows the *parallelogram rule* for vector addition.

Lines

The line in $\mathbb{R}^2$ through $(1, 2)$ and $(3, 1)$ is comprised of the vectors in this set

$$\{\begin{pmatrix}1\\2\end{pmatrix} + t\begin{pmatrix}2\\-1\end{pmatrix} \mid t \in \mathbb{R}\}$$



(that is, it is comprised of the endpoints of those vectors).

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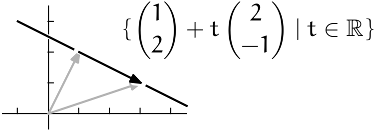


(that is, it is comprised of the endpoints of those vectors). The vector associated with the parameter t

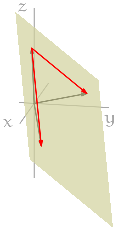
$$\begin{pmatrix}2\\-1\end{pmatrix}$$

is a *direction vector* for the line. Lines in higher dimensions work the same way.

Planes

The plane in $\mathbb{R}^3$ through the points $(1,0,5)$, $(2,1,-3)$, and $(-2,4,0.5)$ consists of (endpoints of) the vectors in this set.

$$\{\begin{pmatrix}1\\0\\5\end{pmatrix}+t\begin{pmatrix}1\\1\\-8\end{pmatrix}+s\begin{pmatrix}-3\\4\\-4.5\end{pmatrix}\mid t,s\in\mathbb{R}\}$$



The column vectors associated with a parameter, shown here in red,

$$\begin{pmatrix}1\\1\\-8\end{pmatrix}=\begin{pmatrix}2\\1\\-3\end{pmatrix}-\begin{pmatrix}1\\0\\5\end{pmatrix}\qquad\begin{pmatrix}-3\\4\\-4.5\end{pmatrix}=\begin{pmatrix}-2\\4\\0.5\end{pmatrix}-\begin{pmatrix}1\\0\\5\end{pmatrix}$$

have their whole body in the plane, not just their endpoint.

A set of the form $\{\vec{p}+t_1\vec{v}_1+t_2\vec{v}_2+\cdots+t_k\vec{v}_k\mid t_1,\ldots,t_k\in\mathbb{R}\}$ where $\vec{v}_1,\ldots,\vec{v}_k\in\mathbb{R}^n$ and $k\leqslant n$ is a *k-dimensional linear surface* (or *k-flat*).

Length and angle measures

Length

?? *Definition* The *length* of a vector $\vec{v} \in \mathbb{R}^n$ is the square root of the sum of the squares of its components.

$$|\vec{v}\,| = \sqrt{v_1^2 + \cdots + v_n^2}$$

Example The length of

$$\vec{v} = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}$$

is $|\vec{v}| = \sqrt{1+4+9} = \sqrt{14}$.

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For any nonzero vector $\vec{v}$, the length one vector with the same direction is $\vec{v}/|\vec{v}|$. We say that this *normalizes* $\vec{v}$ to unit length.

$$\frac{\vec{v}}{|\vec{v}|} = \begin{pmatrix} -1/\sqrt{14} \\ -2/\sqrt{14} \\ -3/\sqrt{14} \end{pmatrix}$$

Note that $\sqrt{(-1/\sqrt{14})^2 + (-2/\sqrt{14})^2 + (-3/\sqrt{14})^2}$ equals 1.

Dot product

?? *Definition* The *dot product* (or *inner product* or *scalar product*) of two n-component real vectors is the linear combination of their components.

$$\vec{u} \bullet \vec{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

We can also use $\mathbf{u}^T \boldsymbol{v}$ to denote inner product.

Example The dot product of two vectors

$$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 4 \end{pmatrix} = 3 - 3 - 4 = -4$$

is a scalar, not a vector.

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is a scalar, not a vector.

The dot product of a vector with itself $\vec{v} \bullet \vec{v} = v_1^2 + \cdots + v_n^2$ is the square of the vector's length.

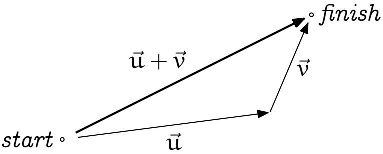
Triangle Inequality

?? *Theorem* For any $\vec{u}, \vec{v} \in \mathbb{R}^n$,

$$|\vec{u} + \vec{v}| \leqslant |\vec{u}| + |\vec{v}|$$

with equality if and only if one of the vectors is a nonnegative scalar multiple of the other one.

This is the source of the familiar saying, “The shortest distance between two points is in a straight line.”



?? *Proof* Since all the numbers are positive, the inequality holds if and only if its square holds.

$$
\begin{aligned}
|\vec{u}+\vec{v}|^2 &\leqslant (\,|\vec{u}\,|+|\vec{v}\,|\,)^2 \\
(\,\vec{u}+\vec{v}\,)\bullet(\,\vec{u}+\vec{v}\,) &\leqslant |\vec{u}\,|^2+2\,|\vec{u}\,|\,|\vec{v}\,|+|\vec{v}\,|^2 \\
\vec{u}\bullet\vec{u}+\vec{u}\bullet\vec{v}+\vec{v}\bullet\vec{u}+\vec{v}\bullet\vec{v} &\leqslant \vec{u}\bullet\vec{u}+2\,|\vec{u}\,|\,|\vec{v}\,|+\vec{v}\bullet\vec{v} \\
2\,\vec{u}\bullet\vec{v} &\leqslant 2\,|\vec{u}\,|\,|\vec{v}\,|
\end{aligned}
$$

?? *Proof* Since all the numbers are positive, the inequality holds if and only if its square holds.

$$\begin{aligned}
|\vec{u}+\vec{v}|^2 &\leqslant (\,|\vec{u}\,|+|\vec{v}\,|\,)^2 \\
(\,\vec{u}+\vec{v}\,)\cdot(\,\vec{u}+\vec{v}\,) &\leqslant |\vec{u}\,|^2+2\,|\vec{u}\,|\,|\vec{v}\,|+|\vec{v}\,|^2 \\
\vec{u}\cdot\vec{u}+\vec{u}\cdot\vec{v}+\vec{v}\cdot\vec{u}+\vec{v}\cdot\vec{v} &\leqslant \vec{u}\cdot\vec{u}+2\,|\vec{u}\,|\,|\vec{v}\,|+\vec{v}\cdot\vec{v} \\
2\,\vec{u}\cdot\vec{v} &\leqslant 2\,|\vec{u}\,|\,|\vec{v}\,|
\end{aligned}$$

That, in turn, holds if and only if the relationship obtained by multiplying both sides by the nonnegative numbers $|\vec{u}\,|$ and $|\vec{v}\,|$

$$2\,(\,|\vec{v}\,|\,\vec{u}\,)\cdot(\,|\vec{u}\,|\,\vec{v}\,) \leqslant 2\,|\vec{u}\,|^2\,|\vec{v}\,|^2$$

and rewriting

$$0 \leqslant |\vec{u}\,|^2\,|\vec{v}\,|^2 - 2\,(\,|\vec{v}\,|\,\vec{u}\,)\cdot(\,|\vec{u}\,|\,\vec{v}\,) + |\vec{u}\,|^2\,|\vec{v}\,|^2$$

is true. But factoring shows that it is true

$$0 \leqslant (\,|\vec{u}\,|\,\vec{v}-|\vec{v}\,|\,\vec{u}\,)\cdot(\,|\vec{u}\,|\,\vec{v}-|\vec{v}\,|\,\vec{u}\,)$$

since it only says that the square of the length of the vector $|\vec{u}\,|\,\vec{v}-|\vec{v}\,|\,\vec{u}$ is not negative.

As for equality, it holds when, and only when, $|\vec{u}\,|\,\vec{v}-|\vec{v}\,|\,\vec{u}$ is $\vec{0}$. The check that $|\vec{u}\,|\,\vec{v}=|\vec{v}\,|\,\vec{u}$ if and only if one vector is a nonnegative real scalar multiple of the other is easy. QED

Cauchy-Schwarz Inequality

?? *Corollary* For any $\vec{u}, \vec{v} \in \mathbb{R}^n$,

$$|\vec{u} \bullet \vec{v}| \leqslant |\vec{u}||\vec{v}|$$

with equality if and only if one vector is a scalar multiple of the other.

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with equality if and only if one vector is a scalar multiple of the other.

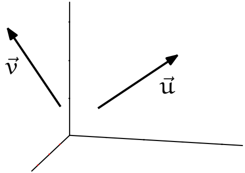
?? *Proof* Exercise.

Angle measure

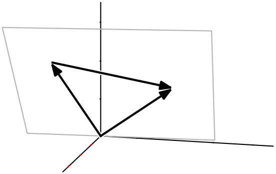
Definition The *angle* between two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ is this.

$$\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}\,|\,|\vec{v}\,|} \right)$$

We motivate that definition with two vectors in $\mathbb{R}^3$.



If neither is a multiple of the other then they determine a plane, because if we put them in canonical position then the origin and the endpoints make three noncolinear points. Consider the triangle formed by $\vec{u}$, $\vec{v}$, and $\vec{u} - \vec{v}$.



Apply the Law of Cosines: $|\vec{u}-\vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2\,|\vec{u}|\,|\vec{v}|\cos\theta$ where θ is the angle that we want to find. The left side gives

$$\begin{aligned}&(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2 \\ &\qquad = (u_1^2 - 2u_1v_1 + v_1^2) + (u_2^2 - 2u_2v_2 + v_2^2) + (u_3^2 - 2u_3v_3 + v_3^2)\end{aligned}$$

while the right side gives this.

$$(u_1^2 + u_2^2 + u_3^2) + (v_1^2 + v_2^2 + v_3^2) - 2\,|\vec{u}|\,|\vec{v}|\cos\theta$$

Canceling squares u_1^2 . . . , v_3^2 and dividing by 2 gives the formula.

Corollary Vectors from $\mathbb{R}^n$ are orthogonal, that is, perpendicular, if and only if their dot product is zero. They are parallel if and only if their dot product equals the product of their lengths.

Definition and examples

Introduction

We have seen that General = Particular + Homogeneous. A particular solution is just a single vector. But the homogeneous part is potentially more interesting because it may contain infinitely many vectors. In this section we will s focus on two properties of such sets.

Here is a typical one, a plane through the origin in $\mathbb{R}^3$.

$$S = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + 2y + 3z = 0 \} = \{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} z \mid y, z \in \mathbb{R} \}$$

Take some vectors from that set, such as these.

$$\begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 1 \\ -4/3 \end{pmatrix} \quad \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}$$

Observe that if we add any two members of S,

$$\begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ -4/3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -5/3 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 2 \\ 1 \\ -4/3 \end{pmatrix} + \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -1/3 \end{pmatrix}$$

then the result is another element of S

Similarly, if we rescale any member of S,

$$3 \cdot \begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} \quad \text{or} \quad -12 \cdot \begin{pmatrix} 2 \\ 1 \\ -4/3 \end{pmatrix} = \begin{pmatrix} -24 \\ -12 \\ -16 \end{pmatrix}$$

then the result is another member of S.

In general, we can take any linear combination of members of S

$$9 \begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 1 \\ -4/3 \end{pmatrix} + \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 2/3 \end{pmatrix}$$

and the result is a member of S.

We will study collections with these properties.

Vector space

?? *Definition* A *vector space* (over $\mathbb{R}$) consists of a set V along with two operations '+' and '$\cdot$' subject to the conditions that for all vectors $\vec{v}, \vec{w}, \vec{u} \in V$, and all *scalars* $r, s \in \mathbb{R}$:

1) the set V is *closed* under vector addition, that is, $\vec{v} + \vec{w} \in V$
2) vector addition is commutative $\vec{v} + \vec{w} = \vec{w} + \vec{v}$
3) vector addition is associative $(\vec{v} + \vec{w}) + \vec{u} = \vec{v} + (\vec{w} + \vec{u})$
4) there is a *zero vector* $\vec{0} \in V$ such that $\vec{v} + \vec{0} = \vec{v}$ for all $\vec{v} \in V$
5) each $\vec{v} \in V$ has an *additive inverse* $\vec{w} \in V$ such that $\vec{w} + \vec{v} = \vec{0}$

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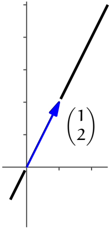
5) each $\vec{v} \in V$ has an *additive inverse* $\vec{w} \in V$ such that $\vec{w} + \vec{v} = \vec{0}$
6) the set V is closed under scalar multiplication, that is, $r \cdot \vec{v} \in V$
7) scalar multiplication distributes over addition of scalars $(r + s) \cdot \vec{v} = r \cdot \vec{v} + s \cdot \vec{v}$
8) scalar multiplication distributes over vector addition $r \cdot (\vec{v} + \vec{w}) = r \cdot \vec{v} + r \cdot \vec{w}$
9) ordinary multiplication of scalars associates with scalar multiplication $(rs) \cdot \vec{v} = r \cdot (s \cdot \vec{v})$
10) multiplication by the scalar 1 is the identity operation $1 \cdot \vec{v} = \vec{v}$.

Example Let V be the line with slope 2 that passes through the origin in the plane.



$$V = \{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid y = 2x \}$$

It is a set consisting of vectors. Here are some of its infinitely many elements.

$$\begin{pmatrix} 4 \\ 8 \end{pmatrix} \quad \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -100 \\ -200 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We will show that this set is a vector space, where the operations are the usual vector addition and scalar multiplication.

We will verify conditions (1)-(10) above.

Before we go through those details, we first reiterate the intuition. A vector space is a place where linear combinations can happen. For instance, this linear combination of vectors from V

$$3 \cdot \begin{pmatrix} 4 \\ 8 \end{pmatrix} + 6 \cdot \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} - (1/10) \cdot \begin{pmatrix} -100 \\ -200 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} \qquad (*)$$

totals to a member of V, a two-tall vector whose second compenent is twice its first.

To say "linear combinations can happen" requires that we have an addition operation and a scalar multiplication. For an operation to deserve to be called an addition it must satisfy some conditions, such as commutativity, and similarly we have some conditions on scalar multiplication.

But the key conditions, as we illustrated in $(*)$ above, are the closure conditions: when you take a combination of vectors from the space then it must total to another vector from the space.

Now we will verify that V is a vector space under the natural addition and scalar multiplication operations.

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} \qquad r \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} rx \\ ry \end{pmatrix}$$

Because this is the first time through the definition we will verify the ten conditions at length.

First is condition (1), closure under addition, that the sum of two members of V is also a member of V.

Take two vectors

$$\vec{v}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

that are members of V, that is, are subject to the restriction that the second component is twice the first: $y_1 = 2x_1$ and $y_2 = 2x_2$. Their sum

$$\vec{v}_1 + \vec{v}_2 = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

is also a member of V because it satisfies the same restriction: $y_1 + y_2 = 2x_1 + 2x_2 = 2(x_1 + x_2)$.

Condition (2), commutativity of addition, is also straightforward. Again we take two vectors

$$\vec{v}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

subject to $y_1 = 2x_1$ and $y_2 = 2x_2$.

The sums in the two orders are

$$\vec{v}_1 + \vec{v}_2 = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} \qquad \vec{v}_2 + \vec{v}_1 = \begin{pmatrix} x_2 + x_1 \\ y_2 + y_1 \end{pmatrix}$$

and the two are equal because $x_1 + x_2 = x_2 + x_1$ and $y_1 + y_2 = y_2 + y_1$, as those are sums of real numbers.

Remark Some conditions, including this one, have nothing to do with the $y = 2x$ restriction. We'll say more about this in the next section.

Condition (3), associativity of addition, is like the prior one. The left side is

$$(\vec{v}_1 + \vec{v}_2) + \vec{v}_3 = \begin{pmatrix} (x_1 + x_2) + x_3 \\ (y_1 + y_2) + y_3 \end{pmatrix}$$

while the right side is

$$\vec{v}_1 + (\vec{v}_2 + \vec{v}_3) = \begin{pmatrix} x_1 + (x_2 + x_3) \\ y_1 + (y_2 + y_3) \end{pmatrix}$$

and the two are equal, since the entries are real numbers.

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and the two are equal, since the entries are real numbers.

Condition (4) asserts that a member of V is an additive identity. We can just exhibit the called-for vector.

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This is a member of V since its second component is twice its first. It is the identity element with respect to addition.

$$\vec{v} + \vec{0} = \begin{pmatrix} x + 0 \\ y + 0 \end{pmatrix} = \vec{v}$$

Condition (5), existence of an additive inverse, is also a matter of producing the desired element. Given

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

subject to $y = 2x$, consider this vector.

$$\vec{w} = \begin{pmatrix} -x \\ -y \end{pmatrix}$$

Note that $\vec{w} \in V$ because $-y = 2(-x)$ follows from $y = 2x$. Note also that the two add to make the zero vector $\vec{v} + \vec{w} = \vec{0}$.

We finish by verifying the five conditions having to do with scalar multiplication.

Condition (6) is closure under scalar multiplication. Consider a scalar $r \in \mathbb{R}$ and a vector $\vec{v} \in V$

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

(that is, such that $y = 2x$). Then

$$r \cdot \vec{v} = \begin{pmatrix} rx \\ ry \end{pmatrix}$$

satisfies the restriction that the second component is twice the first $ry = 2(rx)$, because that equation follows from $y = 2x$. Thus $r\vec{v}$ is also a member of V.

Condition (7) is that real number addition distributes over scalar multiplication. Fix scalars $r, s \in \mathbb{R}$, and a vector $\vec{v} \in V$.

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Then we have this.

$$(r + s) \cdot \vec{v} = \begin{pmatrix} (r+s)x \\ (r+s)y \end{pmatrix} = \begin{pmatrix} rx + sx \\ ry + sy \end{pmatrix} = r\vec{v} + s\vec{v}$$

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For (8), distributivity of vector addition over scalar multiplication, take a scalar $r \in \mathbb{R}$ and two vectors $\vec{v}_1, \vec{v}_2 \in V$.

$$r(\vec{v}_1 + \vec{v}_2) = \begin{pmatrix} r(x_1+x_2) \\ r(y_1+y_2) \end{pmatrix} = \begin{pmatrix} rx_1+rx_2 \\ ry_1+ry_2 \end{pmatrix} = r\vec{v}_1 + r\vec{v}_2$$

For condition (9) suppose $r, s \in \mathbb{R}$ and $\vec{v} \in V$. Compare these two.

$$(rs) \cdot \vec{v} = \begin{pmatrix} (rs)x \\ (rs)y \end{pmatrix} \qquad r \cdot (s\vec{v}) = \begin{pmatrix} r(sx) \\ r(sy) \end{pmatrix}$$

They are equal because the expressions in the components are multiplications of real numbers.

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Condition (10) is simple:

$$1 \cdot \vec{v} = \begin{pmatrix} 1 \cdot x \\ 1 \cdot y \end{pmatrix} = \vec{v}$$

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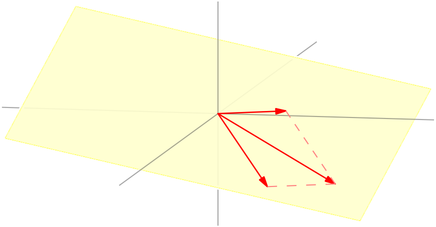
The conclusion: V is a vector space, under the natural addition and scalar multiplication operations.

Example This plane through the origin

$$P = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid 2x + y + 3z = 0 \}$$

is a vector space (under the natural operations). We will verify the two closure conditions (1) and (6); verifying the other conditions is similar to the prior example.

Condition (1) is that two vectors in the plane sum to a vector in the plane. Before we confirm that algebraically, here is the geometry.



For (1), take two members of the plane,

$$\vec{p}_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \vec{p}_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

so that both $2x_1 + y_1 + 3z_1 = 0$ and $2x_2 + y_2 + 3z_2 = 0$. The sum is

$$\vec{p}_1 + \vec{p}_2 = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}$$

and this sum is in the plane because it satisfies the restriction.

$$2(x_1 + x_2) + (y_1 + y_2) + 3(z_1 + z_2) = (2x_1 + y_1 + 3z_1) + (2x_2 + y_2 + 3z_2) = 0$$

For condition (6) take a member of the plane

$$\vec{p} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \qquad 2x + y + 3z = 0$$

and multiply by a scalar $r \in \mathbb{R}$.

$$r\vec{p} = \begin{pmatrix} rx \\ ry \\ rz \end{pmatrix}$$

Then $r\vec{p}$ is a member of the plane P because $2(rx) + (ry) + 3(rz) = r(2x + y + 3z) = 0$.

$\mathbb{R}^n$

The set of n-tall vectors is a vector space under the natural operations. All ten conditions are easy; we will just verify condition (1). Where

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \qquad \vec{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$$

then the sum

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}$$

is also a member of $\mathbb{R}^n$. (There are no restrictions to check, since every n-tall vector is a member of $\mathbb{R}^n$.)

Example Consider the set of quadratic polynomials.

$$\mathcal{P}_2 = \{ a_0 + a_1 x + a_2 x^2 \mid a_0, a_1, a_2 \in \mathbb{R} \}$$

Some members are $3 + 2x + 1x^2$, $10 + 0x + 5x^2$, and $0 + 0x + 0x^2$.

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Some members are $3 + 2x + 1x^2$, $10 + 0x + 5x^2$, and $0 + 0x + 0x^2$. This is a vector space under the usual operations of polynomial addition

$$(a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2$$

and scalar multiplication.

$$r \cdot (a_0 + a_1x + a_2x^2) = (ra_0) + (ra_1)x + (ra_2)x^2$$

Remember the intuition that a vector space is a place where linear combinations can happen. Here is a sample combination in $\mathcal{P}_2$

$$4 \cdot (1 + 2x + 3x^2) - (1/5) \cdot (10 + 5x^2) = 2 + 8x + 11x^2$$

illustrating that a linear combination of quadratic polynomials is a quadratic polynomial.

We won't give a full verification but we will check the closure conditions (1) and (6).

For (1) note that if $\vec{v} = a_0 + a_1 x + a_2 x^2$ and $\vec{w} = (b_0 + b_1 x + b_2 x^2)$ then $\vec{v} + \vec{w} = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2$ is also a quadratic polynomial.

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Similarly, for (6) note that if $\vec{v} = a_0 + a_1 x + a_2 x^2$ then $r \cdot \vec{v} = (ra_0) + (ra_1)x + (ra_2)x^2$ is also a member of $\mathcal{P}_2$.

Example The set of 3×3 matrices

$$\mathcal{M}_{3\times 3} = \{ \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix} \mid a_{i,j} \in \mathbb{R}\}$$

is a vector space under the usual matrix addition and scalar multiplication. The check of the ten conditions is straightforward.

Here is a sample linear combination.

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ -1 & 3 & 1/2 \end{pmatrix} - 3 \begin{pmatrix} 0 & 0 & 2 \\ 1 & 1 & 1 \\ 0 & 4 & 3/2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -5 \\ -1 & -3 & 1 \\ -1 & -9 & -4 \end{pmatrix}$$

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Example The set consisting only of the two-tall zero vector

$$V = \{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \}$$

is a vector space (under the usual vector addition and scalar multiplication operations).

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad r \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

?? *Definition* A one-element vector space is a *trivial* space.

?? *Lemma* In any vector space V, for any $\vec{v} \in V$ and $r \in \mathbb{R}$, we have (1) $0 \cdot \vec{v} = \vec{0}$, (2) $(-1 \cdot \vec{v}) + \vec{v} = \vec{0}$, and (3) $r \cdot \vec{0} = \vec{0}$.

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Proof For (1) note that $\vec{v} = (1+0) \cdot \vec{v} = \vec{v} + (0 \cdot \vec{v})$. Add to both sides the additive inverse of $\vec{v}$, the vector $\vec{w}$ such that $\vec{w} + \vec{v} = \vec{0}$.

$$\begin{aligned}
\vec{w} + \vec{v} &= \vec{w} + \vec{v} + 0 \cdot \vec{v} \\
\vec{0} &= \vec{0} + 0 \cdot \vec{v} \\
\vec{0} &= 0 \cdot \vec{v}
\end{aligned}$$

Item (2) is easy: $(-1 \cdot \vec{v}) + \vec{v} = (-1+1) \cdot \vec{v} = 0 \cdot \vec{v} = \vec{0}$. For (3), $r \cdot \vec{0} = r \cdot (0 \cdot \vec{0}) = (r \cdot 0) \cdot \vec{0} = \vec{0}$ will do. QED

Subspaces and spanning sets

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?? *Definition* For any vector space, a *subspace* is a subset that is itself a vector space, under the inherited operations.

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Example We have seen that in the vector space $\mathbb{R}^2$, the line $y = 2x$

$$S = \{\begin{pmatrix} a \\ 2a \end{pmatrix} \mid a \in \mathbb{R}\} = \{\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot a \mid a \in \mathbb{R}\}$$

is a subspace. The operations are the ones from $\mathbb{R}^2$, as required by the definition above. Earlier, to show it is a vector space we checked the ten conditions from the definition. The result below says that to check if something is a subspace there is an easier way.

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Example This subset of $\mathcal{M}_{2\times2}$ is a subspace.

$$S = \{\begin{pmatrix} a & b \\ a & b \end{pmatrix} \mid a, b \in \mathbb{R}\} = \{\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \cdot a + \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \cdot b \mid a, b \in \mathbb{R}\}$$

Addition and scalar multiplication are the same as in $\mathcal{M}_{2\times2}$.

Example This is not a subspace of $\mathbb{R}^3$.

$$T = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + y + z = 1 \}$$

It is a subset of $\mathbb{R}^3$ but it is not a vector space. One condition that it violates is that it is not closed under vector addition: here are two elements of T that sum to a vector that is not an element of T.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

(Another reason that it is not a vector space is that it does not satisfy condition (6). Still another is that it does not contain the zero vector.)

?? *Lemma* For a nonempty subset S of a vector space, under the inherited operations the following are equivalent statements.

(1) S is a subspace of that vector space

(2) S is closed under linear combinations of pairs of vectors: for any vectors $\vec{s}_1, \vec{s}_2 \in S$ and scalars r_1, r_2 the vector $r_1\vec{s}_1 + r_2\vec{s}_2$ is in S

(3) S is closed under linear combinations of any number of vectors: for any vectors $\vec{s}_1, \ldots, \vec{s}_n \in S$ and scalars $r_1, \ldots, r_n$ the vector $r_1\vec{s}_1 + \cdots + r_n\vec{s}_n$ is an element of S.

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The book has the full proof. For the idea, recall that in the first example of a vector space we remarked that some of the conditions do not depend on the restriction. For instance, there $\vec{v}_1 + \vec{v}_2$ equals $\vec{v}_2 + \vec{v}_1$

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} = \begin{pmatrix} x_2 + x_1 \\ y_2 + y_1 \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

because the vector components are real number sums. Only the closure conditions need verification. Statements (2) and (3) above combine the closure conditions into a single one, to make the verification faster.

Example The vector space of quadratic polynomials $\mathcal{P}_2 = \{a_0 + a_1x + a_2x^2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$ has a subspace comprised of the linear polynomials $L = \{b_0 + b_1x \mid b_0, b_1 \in \mathbb{R}\}$. By the prior result, to verify that we need only check closure under linear combinations of two members.

$$r \cdot (b_0 + b_1x) + s \cdot (c_0 + c_1x) = (rb_0 + sc_0) + (rb_1 + sc_1)x$$

The right side is a linear polynomial with real coefficients, and so is a member of L. Thus L is a subspace of $\mathcal{P}_2$.

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Example Another subspace of $\mathcal{P}_2$ is the set of quadratic polynomials having three equal coefficients.

$$M = \{a + ax + ax^2 \mid a \in \mathbb{R}\} = \{(1 + x + x^2) \cdot a \mid a \in \mathbb{R}\}$$

Verify that it is a subspace by considering a linear combination of two of its members (under the inherited operations).

$$r \cdot (a + ax + ax^2) + s \cdot (b + bx + bx^2) = (ra + sb) + (ra + sb)x + (ra + sb)x^2$$

The result is a quadratic polynomial with three equal coefficients and so M is closed under linear combinations.

The prior subspace example parametrizes the description.

$$M = \{a + ax + ax^2 \mid a \in \mathbb{R}\} = \{(1 + x + x^2) \cdot a \mid a \in \mathbb{R}\}$$

That proves to be a great way to understand vector spaces.

Example This subset of $\mathbb{R}^3$ is a plane.

$$P = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid 2x - y + z = 0\}$$

We could verify that it is a subspace by checking that it is closed under linear combination, as above. However, that's easier if we first parametrize.

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$$P = \{ \begin{pmatrix} (1/2)y - (1/2)z \\ y \\ z \end{pmatrix} \mid y, z \in \mathbb{R}\} = \{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} z \mid y, z \in \mathbb{R}\}$$

With the parametrized description

$$P = \{y \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \mid y, z \in \mathbb{R}\} \qquad (*)$$

showing the subspace is closed under linear combinations is straightforward (here, $r_1, r_2 \in \mathbb{R}$).

$$\begin{aligned} r_1 \cdot \bigl(y_1 \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} + z_1 \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}\bigr) + r_2 \cdot \bigl(y_2 \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} + z_2 \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix}\bigr) \\ = (r_1 y_1 + r_2 y_2) \cdot \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} + (r_1 z_1 + r_2 z_2) \cdot \begin{pmatrix} -1/2 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

Line $(*)$ describes each member of P as a linear combination of the two vectors. Verifying that P is closed then just involves taking a linear combination of linear combinations. Of course, that gives a linear combination.

Example To show that this is a subspace of $\mathcal{M}_{2\times 2}$

$$Q = \{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a - b = 0 \text{ and } b - c = 0 \}$$

treat that as a two-equation linear system and parametrize. The leading variables are a and b, with free variables c and d.

$$Q = \{ \begin{pmatrix} c & c \\ c & d \end{pmatrix} \mid c, d \in \mathbb{R} \} = \{ c \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mid c, d \in \mathbb{R} \}$$

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$$Q = \{ \begin{pmatrix} c & c \\ c & d \end{pmatrix} \mid c, d \in \mathbb{R} \} = \{ c \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mid c, d \in \mathbb{R} \}$$

To show Q is a subspace, use the lemma's clause (2) by finding that a linear combination of such matrices is such a matrix.

$$\begin{aligned} r_1 \cdot \left(c_1 \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + d_1 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) + r_2 \cdot \left(c_2 \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + d_2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) \\ = (r_1 c_1 + r_2 c_2) \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + (r_1 d_1 + r_2 d_2)) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Span

?? *Definition* The *span* (or *linear closure*) of a nonempty subset S of a vector space is the set of all linear combinations of vectors from S.

$$[S] = \{c_1\vec{s}_1 + \cdots + c_n\vec{s}_n \mid c_1, \ldots, c_n \in \mathbb{R} \text{ and } \vec{s}_1, \ldots, \vec{s}_n \in S\}$$

The span of the empty subset of a vector space is its trivial subspace.

No notation for the span is completely standard. The square brackets used here are common but so are ‘span(S)’ and ‘sp(S)’.

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No notation for the span is completely standard. The square brackets used here are common but so are 'span(S)' and 'sp(S)'.

Example Inside the vector space of all two-wide row vectors, the span of this one-element set

$$S = \{\begin{pmatrix}1 & 2\end{pmatrix}\}$$

is this.

$$[S] = \{\begin{pmatrix}a & 2a\end{pmatrix} \mid a \in \mathbb{R}\} = \{\begin{pmatrix}1 & 2\end{pmatrix}a \mid a \in \mathbb{R}\}$$

Example This is a subset of $\mathbb{R}^3$.

$$\hat{S} = \{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \}$$

Any vector in the xy-plane is a member of the span $[S]$ because any such vector is a combination of the two; for instance, this system has a solution

$$\begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} c_1 + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} c_2$$

(the top two rows gives a linear system with a unique solution).

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(the top two rows gives a linear system with a unique solution). But vectors not in the xy-plane are not in the span. For instance, this system does not have a solution.

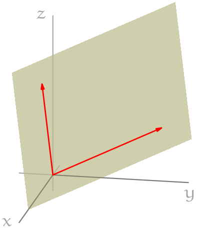
$$\begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} c_1 + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} c_2$$

Example Here is another subset of $\mathbb{R}^3$.

$$P = \{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \}$$

Vectors in the span $[P]$ are combinations of the two, shown in red. The span is a plane through the origin.

$$[P] = \{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \cdot s + \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \cdot t \mid s, t \in \mathbb{R} \}$$



?? *Lemma* In a vector space, the span of any subset is a subspace.

?? *Lemma* In a vector space, the span of any subset is a subspace.

Proof If the subset S is empty then by definition its span is the trivial subspace. If S is not empty then by Lemma **??** we need only check that the span $[S]$ is closed under linear combinations of pairs of elements. For a pair of vectors from that span, $\vec{v} = c_1\vec{s}_1 + \cdots + c_n\vec{s}_n$ and $\vec{w} = c_{n+1}\vec{s}_{n+1} + \cdots + c_m\vec{s}_m$, a linear combination

$$\begin{aligned} p\cdot(c_1\vec{s}_1 + \cdots + c_n\vec{s}_n) + r\cdot(c_{n+1}\vec{s}_{n+1} + \cdots + c_m\vec{s}_m) \\ = pc_1\vec{s}_1 + \cdots + pc_n\vec{s}_n + rc_{n+1}\vec{s}_{n+1} + \cdots + rc_m\vec{s}_m \end{aligned}$$

is a linear combination of elements of S and so is an element of $[S]$ (possibly some of the $\vec{s}_i$'s from $\vec{v}$ equal some of the $\vec{s}_j$'s from $\vec{w}$ but that does not matter). QED

Example This illustrates that a span is closed under linear combinations. Where

$$\hat{S} = \{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \} \subset \mathbb{R}^3$$

these are two elements of the span $[\hat{S}]$.

$$\vec{v}_1 = 5 \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \qquad \vec{v}_2 = -2 \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 10 \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The linear combination of those two $-3\vec{v}_1 + 7\vec{v}_2$ makes another element of the span.

$$-3 \cdot \left(5 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) + 7 \cdot \left(-2 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 10 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) = -29 \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 61 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

This is just an instance of the Linear Combination Lemma: a linear combination of linear combinations is a linear combination.

Spaces and their subspaces

Subspaces of $\mathbb{R}^3$

The next slide shows a sample of subspaces of the vector space $\mathbb{R}^3$: the entire space, planes, lines, and the trivial subspace. Subsets are drawn connected to supersets, on the levels directly above and below.

On the level one up from the bottom, the subspaces are lines (in the second and third case, because the conjunction of the two conditions means that each is the intersection of two planes). That is, they are one-dimensional.

On the next level up, the subspaces are planes, two-dimensional. On the top level, the space is 3-space.

$\mathbb{R}^3$

xy-plane $\quad \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + y + z = 0 \} \quad \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + 2z = 0 \} \quad \cdots$

y-axis $\quad \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x - y + z = 0 \text{ and } x + 2z = 0 \} \quad \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid y = 2x \text{ and } z = 0 \} \quad \cdots$

trivial subspace $\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$

Express subspaces as spans

Example This is a subset of $\mathbb{R}^3$.

$$P = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + y + z = 0 \}$$

Here are three members of this set, and an equation showing that the third is a linear combination of the first two, illustrating that it is a vector space.

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} -2 \\ -5 \\ 7 \end{pmatrix}, \vec{v}_3 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \qquad 3\vec{v}_1 + \vec{v}_2 = \vec{v}_3$$

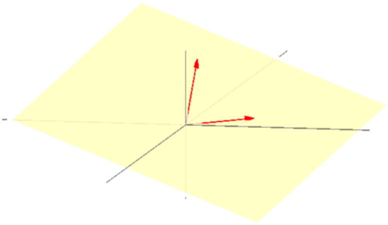
To get a more useful description, take the condition $x + y + z = 0$ to be a one-equation linear system and parametrize.

$$P = \{ \begin{pmatrix} -y - z \\ y \\ z \end{pmatrix} \mid y, z \in \mathbb{R} \} = \{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} z \mid y, z \in \mathbb{R} \}$$

Thus the plane is the span of those two vectors.

The subspace $P \subseteq \mathbb{R}^3$ is a plane passing through the origin.

$$P = [\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \}]$$



The two vectors from the spanning set

$$\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

are in red. For each, its entire body lies in the plane.

Example For the plane

$$\hat{P} = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + 2z = 0 \}$$

repeat the process

$$\hat{P} = \{ \begin{pmatrix} -2z \\ y \\ z \end{pmatrix} \mid y, z \in \mathbb{R} \} = \{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} z \mid y, z \in \mathbb{R} \}$$

to express it as a span.

$$\hat{P} = [\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \}]$$

Example For the plane

$$\hat{P} = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + 2z = 0 \}$$

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to express it as a span.

$$\hat{P} = [\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \}]$$

Example The xy-plane is a span in a natural way.

$$xy \text{ plane} = [\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \}]$$

Example Next we parametrize the lines. The conditions in the set description

$$L = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x - y + z = 0 \text{ and } x + 2z = 0 \}$$

make a linear system.

$$\begin{array}{rcl} x - y + z = 0 \\ x \phantom{{}-y} + 2z = 0 \end{array} \xrightarrow{-\rho_1+\rho_2} \begin{array}{rcl} x - y + z = 0 \\ y + z = 0 \end{array}$$

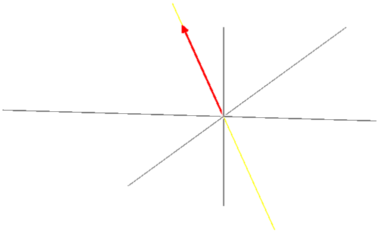
Parametrizing

$$L = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid y = -z \text{ and } x = -2z \}$$

gives this.

$$L = \{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} z \mid z \in \mathbb{R} \} = [\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \}]$$

Here the line, the subspace, is in yellow.



In red is the vector used in the description.

$$\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

Its endpoint lies behind the plane of the screen, in the octant where x and y are negative and z is positive.

Example The line

$$\hat{L} = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid y = 2x \text{ and } z = 0 \}$$

is easy to describe in a parametrized way.

$$\begin{aligned} \hat{L} &= \{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} y \mid y \in \mathbb{R} \} \\ &= [\{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \}] \end{aligned}$$

Example The line

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is easy to describe in a parametrized way.

$$\begin{aligned} \hat{L} &= \{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} y \mid y \in \mathbb{R} \} \\ &= [\{ \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} \}] \end{aligned}$$

Example The y-axis is also easy to describe as a span.

$$y\text{-axis} = [\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \}]$$

Example We can describe the entire space as a span.

$$\mathbb{R}^3 = [\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \}]$$

Example We can describe the entire space as a span.

$$\mathbb{R}^3 = [\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \}]$$

Example We can do the same for the trivial subspace.

$$\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \} = [\{\}]$$

(Remember that a sum of zero-many vectors is the zero vector.)

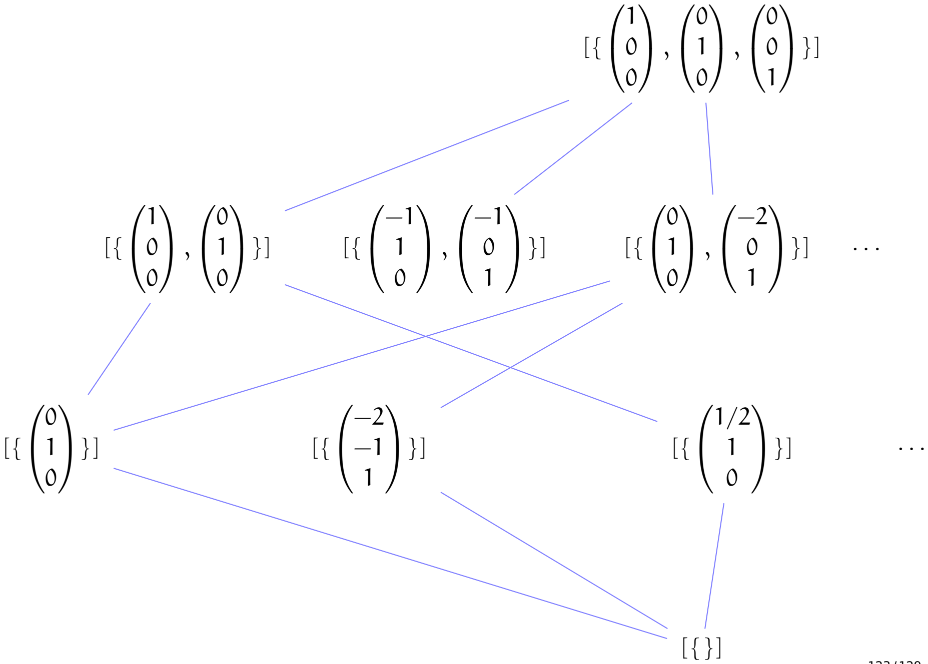
$\mathbb{R}^3$'s diagram reprised

The following slide repeats the diagram of $\mathbb{R}^3$'s subspaces, showing the same subspaces. On this diagram the same subspaces they are described as spans, where the spanning set uses a minimal number of vectors.

By 'minimal' we mean: while we could describe the xy-plane in either of these ways,

$$xy\text{-plane} = [\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\}] = [\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}2\\-1\\0\end{pmatrix}\}]$$

the second has an extra vector, so the next slide doesn't use that description. (We will soon make this precise.)


$[\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\end{pmatrix}\}]$
$[\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix}\}]$
$[\{\begin{pmatrix}-1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\}]$
$[\{\begin{pmatrix}0\\1\\0\end{pmatrix},\begin{pmatrix}-2\\0\\1\end{pmatrix}\}]$
$\ldots$
$[\{\begin{pmatrix}0\\1\\0\end{pmatrix}\}]$
$[\{\begin{pmatrix}-2\\-1\\1\end{pmatrix}\}]$
$[\{\begin{pmatrix}1/2\\1\\0\end{pmatrix}\}]$
$\ldots$
$[\{\}]$

$\mathcal{P}_2$'s subspaces

The next slide has a picture of some of the subspaces of the space of quadratic polynomials. As with the $\mathbb{R}^3$ diagram, subsets are shown connected to supersets on the adjacent level.

With $\mathbb{R}^3$ the geometry gave us a good start for a natural classification of subspaces into planes, lines, etc. In this space there is less of that sense. But there are a couple of natural subspaces.

$$\text{linear polynomials} = \{0x^2 + bx + c \mid b, c \in \mathbb{R}\}$$
$$\text{constant polynomials} = \{0x^2 + 0x + c \mid c \in \mathbb{R}\}$$

These are on the next slide, along with a couple of more-generic spaces.

$\mathcal{P}_2$

linear: $\{\, bx + c \mid b, c \in \mathbb{R} \,\}$ $\qquad$ $\{\, ax^2 + bx + c \mid a + b + c = 0 \,\}$ $\qquad \cdots$

constant: $\{\, c \mid c \in \mathbb{R} \,\}$ $\qquad$ $\{\, ax^2 + bx + c \mid a - c = 0 \text{ and } b - c = 0 \,\}$ $\qquad \cdots$

trivial subspace

Parametrizing

Example For the top level, this will do.

$$\mathcal{P}_2 = [\{x^2, x, 1\}] = [\{x^2 + 0x + 0, 0x^2 + x + 0, 0x^2 + 0x + 1\}]$$

Example On the bottom level, the trivial subspace is the span of the empty set.

$$\{0\} = \{0x^2 + 0x + 0\} = [\{\,\}]$$

Parametrizing

Example For the top level, this will do.

$$\mathcal{P}_2 = [\{x^2, x, 1\}] = [\{x^2 + 0x + 0, 0x^2 + x + 0, 0x^2 + 0x + 1\}]$$

Example On the bottom level, the trivial subspace is the span of the empty set.

$$\{0\} = \{0x^2 + 0x + 0\} = [\{\,\}]$$

Example The set of linear polynomials has a natural expression as a span.

$$\{bx + c \mid b, c \in \mathbb{R}\} = [\{x, 1\}]$$

Example The set of constant polynomials is similar.

$$\{0x^2 + 0x + c \mid c \in \mathbb{R}\} = [\{1\}]$$

Example Parametrize $P = \{ax^2 + bx + c \mid a + b + c = 0\}$ in much the same way as for subspaces of real three-space: treat the restriction as a one-equation linear system and parametrize.

$$\begin{aligned} P &= \{ax^2 + bx + c \mid a = -b - c\} \\ &= \{(-b - c) \cdot x^2 + bx + c \mid b, c \in \mathbb{R}\} \\ &= \{(-x^2 + x) \cdot b + (-x^2 + 1) \cdot c \mid b, c \in \mathbb{R}\} \end{aligned}$$

The spanning set is $\{-x^2 + x, -x^2 + 1\}$.

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The spanning set is $\{-x^2 + x, -x^2 + 1\}$.

Example Simliarly for $\hat{P} = \{ax^2 + bx + c \mid a - c = 0 \text{ and } b - c = 0\}$.

$$\begin{aligned} \hat{P} &= \{ax^2 + bx + c \mid a = c \text{ and } b = c\} \\ &= \{cx^2 + cx + c \mid c \in \mathbb{R}\} \\ &= \{(x^2 + x + 1) \cdot c \mid c \in \mathbb{R}\} \end{aligned}$$

The spanning set is $\{x^2 + x + 1\}$.

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The spanning set is $\{-x^2 + x, -x^2 + 1\}$.

Example Simliarly for $\hat{P} = \{ax^2 + bx + c \mid a - c = 0 \text{ and } b - c = 0\}$.

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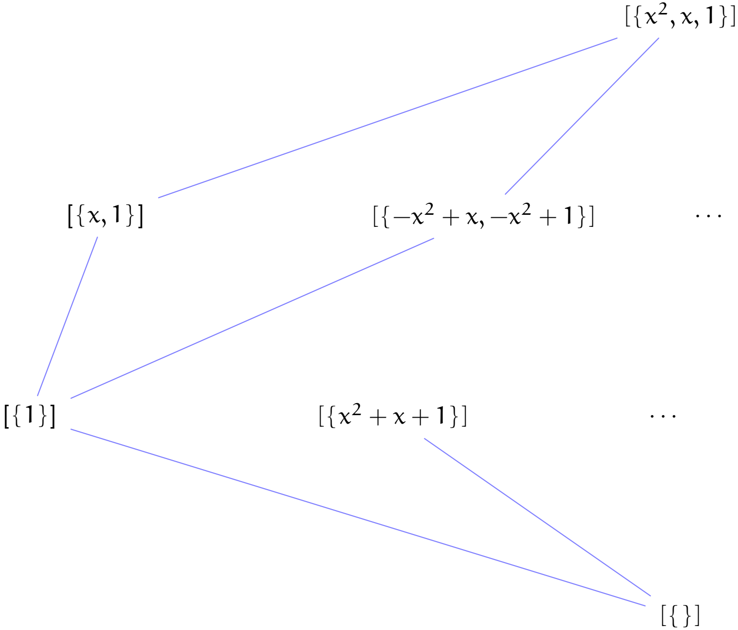
The spanning set is $\{x^2 + x + 1\}$.

The next slide reprises $\mathcal{P}_2$'s diagram, with the subspaces described as spans.


$[\{x^2, x, 1\}]$
$[\{x, 1\}]$
$[\{-x^2+x, -x^2+1\}]$
$\cdots$
$[\{1\}]$
$[\{x^2+x+1\}]$
$\cdots$
$[\{\}]$

Summary

Subspaces are naturally described as spans. In both examples these spans fall naturally into levels, according to the number of elements in a minimal spanning set.

The book's next section gives a precise definition of when a spanning set is 'minimal'. The section after that shows that for any space, two minimal spanning sets have the same number of vectors.